

Strategy use analyses

This vignette contains a full breakdown of the analyses of the self-reported (using Likert scales) mental strategies that participants used to solve the reasoning problems. The most interesting results are synthetically presented in the main manuscript (preprint here). The present document aims to provide a full account of the steps taken to reach these results, and to provide additional analyses that were not included in the manuscript for brevity.

```
library(aphantasiaReasoningViie)
#> Welcome to aphantasiaReasoningViie.
```

Data preparation

First, let's get the clean, analysis-ready data and create cognitive style clusters using OSIVQ scores (see vignette("preparing_data") and vignette("osivq_clusters") for details).

```
df_survey <- get_clean_data()$df_survey
# Clustering OSIVQ data
clustering <- cluster_osivq(df_survey)
# Adding named clusters to the survey data
df_survey <- add_named_clusters(
  df_survey, clustering,
  names = c("Spatialiser", "Visualiser", "Verbaliser"),
  levels = c("Visualiser", "Spatialiser", "Verbaliser"),
  contrasts = c("_visualiser", "_spatialiser", "_verbaliser"),
  base = 1
)
```

By default, the `df_survey` data frame has one variable (one column) containing the responses of each participant for each strategy question (Visual, Spatial, Verbal, Semantic, Sensorimotor). We needed to have all of these five modalities in a single “strategy” variable along with the associated “score” variable for modelling, so we gathered these five columns into two long columns by “pivoting” the strategies data in a long format (reducing the number of columns and increasing the number of rows). This operation is performed by the `pivot_strategies_longer()` function.

```
df_strats_long <- pivot_strategies_longer(df_survey)

dplyr::glimpse(df_strats_long)
#> Rows: 520
#> Columns: 11
#> $ id          <fct> acdn247721443631359lzb, acdn247721443631359lzb, acdn2~
#> $ language    <fct> fr, fr, fr, fr, fr, fr, fr, fr, fr, fr, fr, fr, fr, fr,~
#> $ age         <int> 24, 24, 24, 24, 24, 26, 26, 26, 26, 26, 23, 23, 23,~
#> $ gender      <fct> f, f, f, f, f, f, f, f, f, f, m, m, m, m, f, f, f,~
#> $ group       <fct> Typical, Typical, Typical, Typical, Typical, Aphantasia~
#> $ group_2     <fct> Typical, Typical, Typical, Typical, Typical, Aphantasia~
#> $ group_3     <fct> Typical, Typical, Typical, Typical, Typical, Aphantasia~
#> $ strategy_group <fct> No visual strategy, No visual strategy, No visual strat~
#> $ cluster     <fct> Visualiser, Visualiser, Visualiser, Visualiser, Visuali~
```

```
#> $ strategy      <fct> Visual, Verbal, Spatial, Semantic, Sensorimotor, Visual~  
#> $ score          <fct> no_use, mainly_this_strat, secondary_strat, no_use, onl~
```

Method

As described in the manuscript:

Ordinal cumulative link regression models were fitted using the ordinal package (Christensen, 2023) to predict the score (on a question about the use of a given strategy) with a grouping variable (VVIQ groups, OSIVQ clusters), Strategy (visual, verbal, spatial, semantic or sensorimotor) and their two-way interaction as fixed categorical predictors. We planned to analyse the contrasts between groups for each strategy separately.

Let's break this down.

The grouping variables

We used several grouping variables to classify participants, all of which are in the `df_strats_long` data frame:

- **group** is the 4-group VVIQ classification with an “aphantasia” group ($VVIQ = 16$), “hypophantasia” group ($VVIQ \in [17, 32]$), “typical” group ($VVIQ \in [33, 74]$), and “hyperphantasia” group ($VVIQ \in [75, 80]$). It was not used in the analyses because the hyperphantasia group was too small ($N = 4$).
- **group_3** is the same as **group**, but with the hyperphantasia group merged with the typical group. It is referred to as “VVIQ 3 groups” in the manuscript results.
- **group_2** is the binary classification often used in aphantasia literature that additionally merges the hypophantasia group with the aphantasia group. It is referred to as “VVIQ 2 groups” in the manuscript results.
- **cluster** is the cognitive style classification obtained from clustering OSIVQ scores. It contains three groups: “Visualiser”, “Spatialiser”, and “Verbaliser”. It is referred to as “OSIVQ 3 clusters” in the manuscript results.

The same modelling pipeline was therefore applied three times, once for each of the last three grouping variables.

The modelling pipeline

The ordinal models were fitted using the `fit_clm()` function, which is a wrapper around the `ordinal::clm()` function that simply sets a probit link by default. After model fit, we checked for potential singularity issues and model performance using the `get_singularity()` and `get_performance()` functions created for the occasion (which are convenient wrappers around the *performance* package).

Finally, we tested our hypotheses with marginal contrasts. This task was performed with the `report_contrast()` function, which is a wrapper around several functions from the *emmeans* package. We used it to compute strategy use contrasts between groups for each strategy, contrasts between strategies within each group, and interaction contrasts (differences in strategy contrasts between groups).

Here we go!

Results

VVIQ 2 groups

```
m_strats_vviq_2 <- fit_clm(score ~ group_2 * strategy, df_strats_long)

# Singularity
m_strats_vviq_2 |> get_singularity()
#> The model is not singular, parameter estimates are trustworthy.

# Performance
m_strats_vviq_2 |> get_performance() |> knitr::kable(align = "c")
```

AIC	BIC	RMSE
1268.917	1324.216	2.400

```
# Group contrasts by strategy
m_strats_vviq_2 |> report_contrast(~ group_2 | strategy) |> knitr::kable()
```

Contrast	Strategy	Difference	95% CI	p.value
Aphantasia - Typical	Visual	-1.689	[-2.18, -1.2]	0.000
Aphantasia - Typical	Spatial	-0.397	[-0.82, 0.03]	0.068
Aphantasia - Typical	Verbal	0.267	[-0.14, 0.68]	0.203
Aphantasia - Typical	Semantic	0.229	[-0.28, 0.74]	0.379
Aphantasia - Typical	Sensorimotor	-0.137	[-0.57, 0.29]	0.531

```
# Strategy contrasts within groups
m_strats_vviq_2 |> report_contrast(~ strategy | group_2) |> knitr::kable()
```

Contrast	group_2	Difference	95% CI	p.value
Visual - Spatial	Aphantasia	-1.078	[-1.8, -0.36]	0.000
Visual - Verbal	Aphantasia	-2.454	[-3.18, -1.73]	0.000
Visual - Semantic	Aphantasia	-0.302	[-1.06, 0.46]	0.814
Visual - Sensorimotor	Aphantasia	-0.953	[-1.67, -0.24]	0.003
Spatial - Verbal	Aphantasia	-1.376	[-2, -0.75]	0.000
Spatial - Semantic	Aphantasia	0.775	[0.09, 1.46]	0.017
Spatial - Sensorimotor	Aphantasia	0.125	[-0.51, 0.76]	0.983
Verbal - Semantic	Aphantasia	2.152	[1.47, 2.84]	0.000
Verbal - Sensorimotor	Aphantasia	1.501	[0.87, 2.13]	0.000
Semantic - Sensorimotor	Aphantasia	-0.651	[-1.33, 0.03]	0.069
Visual - Spatial	Typical	0.214	[-0.33, 0.76]	0.822
Visual - Verbal	Typical	-0.498	[-1.04, 0.04]	0.088
Visual - Semantic	Typical	1.616	[0.98, 2.25]	0.000
Visual - Sensorimotor	Typical	0.599	[0.05, 1.15]	0.026
Spatial - Verbal	Typical	-0.713	[-1.26, -0.17]	0.004
Spatial - Semantic	Typical	1.401	[0.77, 2.03]	0.000
Spatial - Sensorimotor	Typical	0.385	[-0.17, 0.94]	0.323
Verbal - Semantic	Typical	2.114	[1.48, 2.75]	0.000
Verbal - Sensorimotor	Typical	1.097	[0.54, 1.65]	0.000
Semantic - Sensorimotor	Typical	-1.017	[-1.65, -0.38]	0.000

```
# Interaction contrasts
```

```
m_strats_vviq_2 |> report_contrast(~ group_2 * strategy, interaction = TRUE) |>  
  knitr::kable()
```

group_2_pairwise	Strategy contrast	Difference	95% CI	p.value
Aphantasia - Typical	Visual - Spatial	-1.292	[-1.94, -0.64]	0.000
Aphantasia - Typical	Visual - Verbal	-1.956	[-2.6, -1.31]	0.000
Aphantasia - Typical	Visual - Semantic	-1.918	[-2.63, -1.21]	0.000
Aphantasia - Typical	Visual - Sensorimotor	-1.552	[-2.2, -0.9]	0.000
Aphantasia - Typical	Spatial - Verbal	-0.664	[-1.26, -0.07]	0.028
Aphantasia - Typical	Spatial - Semantic	-0.626	[-1.29, 0.04]	0.065
Aphantasia - Typical	Spatial - Sensorimotor	-0.260	[-0.86, 0.34]	0.399
Aphantasia - Typical	Verbal - Semantic	0.038	[-0.62, 0.69]	0.910
Aphantasia - Typical	Verbal - Sensorimotor	0.404	[-0.19, 1]	0.182
Aphantasia - Typical	Semantic - Sensorimotor	0.366	[-0.3, 1.03]	0.281

VVIQ 3 groups

```
m_strats_vviq_3 <- fit_clm(score ~ group_3 * strategy, df_strats_long)
```

```
m_strats_vviq_3 |> get_singularity()
```

```
#> The model is not singular, parameter estimates are trustworthy.
```

```
m_strats_vviq_3 |> get_performance() |> knitr::kable(align = "c")
```

AIC	BIC	RMSE
1263.729	1340.298	2.399

```
m_strats_vviq_3 |> report_contrast(~ group_3 | strategy) |> knitr::kable()
```

Contrast	Strategy	Difference	95% CI	p.value
Aphantasia - Hypophantasia	Visual	-1.606	[-2.84, -0.37]	0.006
Aphantasia - Typical	Visual	-2.643	[-3.74, -1.54]	0.000
Hypophantasia - Typical	Visual	-1.036	[-1.77, -0.3]	0.003
Aphantasia - Hypophantasia	Spatial	0.598	[-0.23, 1.42]	0.205
Aphantasia - Typical	Spatial	-0.197	[-0.77, 0.38]	0.703
Hypophantasia - Typical	Spatial	-0.795	[-1.55, -0.04]	0.036
Aphantasia - Hypophantasia	Verbal	-0.093	[-0.85, 0.67]	0.956
Aphantasia - Typical	Verbal	0.234	[-0.33, 0.8]	0.595
Hypophantasia - Typical	Verbal	0.327	[-0.36, 1.02]	0.507
Aphantasia - Hypophantasia	Semantic	-0.202	[-1.11, 0.7]	0.860
Aphantasia - Typical	Semantic	0.154	[-0.55, 0.86]	0.866
Hypophantasia - Typical	Semantic	0.356	[-0.47, 1.18]	0.570
Aphantasia - Hypophantasia	Sensorimotor	0.004	[-0.8, 0.8]	1.000
Aphantasia - Typical	Sensorimotor	-0.137	[-0.72, 0.45]	0.849
Hypophantasia - Typical	Sensorimotor	-0.140	[-0.87, 0.59]	0.893

```
m_strats_vviq_3 |> report_contrast(~ strategy | group_3) |> knitr::kable()
```

Contrast	group_3	Difference	95% CI	p.value
Visual - Spatial	Aphantasia	-2.230	[-3.57, -0.89]	0.000
Visual - Verbal	Aphantasia	-3.378	[-4.72, -2.04]	0.000
Visual - Semantic	Aphantasia	-1.173	[-2.56, 0.21]	0.140
Visual - Sensorimotor	Aphantasia	-1.904	[-3.24, -0.56]	0.001
Spatial - Verbal	Aphantasia	-1.148	[-1.92, -0.38]	0.000
Spatial - Semantic	Aphantasia	1.057	[0.2, 1.91]	0.007
Spatial - Sensorimotor	Aphantasia	0.326	[-0.46, 1.11]	0.787
Verbal - Semantic	Aphantasia	2.205	[1.34, 3.07]	0.000
Verbal - Sensorimotor	Aphantasia	1.474	[0.69, 2.26]	0.000
Semantic - Sensorimotor	Aphantasia	-0.731	[-1.59, 0.13]	0.140
Visual - Spatial	Hypophantasia	-0.025	[-1.12, 1.07]	1.000
Visual - Verbal	Hypophantasia	-1.865	[-2.91, -0.82]	0.000
Visual - Semantic	Hypophantasia	0.231	[-0.89, 1.35]	0.980
Visual - Sensorimotor	Hypophantasia	-0.294	[-1.36, 0.77]	0.944
Spatial - Verbal	Hypophantasia	-1.839	[-2.91, -0.77]	0.000
Spatial - Semantic	Hypophantasia	0.257	[-0.88, 1.4]	0.973
Spatial - Sensorimotor	Hypophantasia	-0.268	[-1.35, 0.82]	0.962
Verbal - Semantic	Hypophantasia	2.096	[1, 3.2]	0.000
Verbal - Sensorimotor	Hypophantasia	1.571	[0.53, 2.61]	0.000
Semantic - Sensorimotor	Hypophantasia	-0.525	[-1.64, 0.59]	0.699
Visual - Spatial	Typical	0.216	[-0.33, 0.76]	0.818
Visual - Verbal	Typical	-0.501	[-1.04, 0.04]	0.085
Visual - Semantic	Typical	1.624	[0.99, 2.25]	0.000
Visual - Sensorimotor	Typical	0.602	[0.05, 1.15]	0.024
Spatial - Verbal	Typical	-0.717	[-1.26, -0.17]	0.003
Spatial - Semantic	Typical	1.408	[0.78, 2.04]	0.000
Spatial - Sensorimotor	Typical	0.386	[-0.17, 0.94]	0.319
Verbal - Semantic	Typical	2.125	[1.49, 2.76]	0.000
Verbal - Sensorimotor	Typical	1.103	[0.55, 1.66]	0.000
Semantic - Sensorimotor	Typical	-1.021	[-1.65, -0.39]	0.000

```
m_strats_vviq_3 |> report_contrast(~ group_3 * strategy, interaction = TRUE) |>
  knitr::kable()
```

group_3_pairwise	Strategy contrast	Difference	95% CI	p.value
Aphantasia - Hypophantasia	Visual - Spatial	-2.204	[-3.44, -0.96]	0.000
Aphantasia - Typical	Visual - Spatial	-2.446	[-3.48, -1.41]	0.000
Hypophantasia - Typical	Visual - Spatial	-0.241	[-1.12, 0.64]	0.590
Aphantasia - Hypophantasia	Visual - Verbal	-1.513	[-2.72, -0.3]	0.014
Aphantasia - Typical	Visual - Verbal	-2.877	[-3.91, -1.84]	0.000
Hypophantasia - Typical	Visual - Verbal	-1.363	[-2.21, -0.52]	0.002
Aphantasia - Hypophantasia	Visual - Semantic	-1.404	[-2.68, -0.13]	0.031
Aphantasia - Typical	Visual - Semantic	-2.796	[-3.89, -1.7]	0.000
Hypophantasia - Typical	Visual - Semantic	-1.392	[-2.32, -0.47]	0.003
Aphantasia - Hypophantasia	Visual - Sensorimotor	-1.610	[-2.84, -0.38]	0.010
Aphantasia - Typical	Visual - Sensorimotor	-2.506	[-3.55, -1.46]	0.000
Hypophantasia - Typical	Visual - Sensorimotor	-0.896	[-1.76, -0.03]	0.042
Aphantasia - Hypophantasia	Spatial - Verbal	0.691	[-0.25, 1.63]	0.149

group_3_pairwise	Strategy contrast	Difference	95% CI	p.value
Aphantasia - Typical	Spatial - Verbal	-0.431	[-1.11, 0.24]	0.211
Hypophantasia - Typical	Spatial - Verbal	-1.122	[-1.98, -0.27]	0.010
Aphantasia - Hypophantasia	Spatial - Semantic	0.800	[-0.22, 1.82]	0.126
Aphantasia - Typical	Spatial - Semantic	-0.351	[-1.11, 0.41]	0.367
Hypophantasia - Typical	Spatial - Semantic	-1.151	[-2.09, -0.21]	0.016
Aphantasia - Hypophantasia	Spatial - Sensorimotor	0.594	[-0.37, 1.56]	0.226
Aphantasia - Typical	Spatial - Sensorimotor	-0.061	[-0.75, 0.63]	0.863
Hypophantasia - Typical	Spatial - Sensorimotor	-0.655	[-1.53, 0.22]	0.143
Aphantasia - Hypophantasia	Verbal - Semantic	0.109	[-0.88, 1.1]	0.829
Aphantasia - Typical	Verbal - Semantic	0.080	[-0.67, 0.83]	0.835
Hypophantasia - Typical	Verbal - Semantic	-0.029	[-0.93, 0.87]	0.950
Aphantasia - Hypophantasia	Verbal - Sensorimotor	-0.097	[-1.02, 0.83]	0.837
Aphantasia - Typical	Verbal - Sensorimotor	0.371	[-0.31, 1.05]	0.286
Hypophantasia - Typical	Verbal - Sensorimotor	0.467	[-0.37, 1.3]	0.274
Aphantasia - Hypophantasia	Semantic - Sensorimotor	-0.206	[-1.22, 0.8]	0.690
Aphantasia - Typical	Semantic - Sensorimotor	0.290	[-0.48, 1.06]	0.458
Hypophantasia - Typical	Semantic - Sensorimotor	0.496	[-0.42, 1.42]	0.290

OSIVQ 3 clusters

```
m_strats_osivq <- fit_clm(score ~ cluster * strategy, df_strats_long)

m_strats_osivq |> get_singularity()
#> The model is not singular, parameter estimates are trustworthy.

m_strats_osivq |> get_performance() |> knitr::kable(align = "c")
```

AIC	BIC	RMSE
1291.258	1367.826	2.403

```
m_strats_osivq |> report_contrast(~ cluster | strategy) |> knitr::kable()
```

Contrast	Strategy	Difference	95% CI	p.value
Visualiser - Spatialiser	Visual	0.810	[0.09, 1.53]	0.024
Visualiser - Verbaliser	Visual	1.521	[0.91, 2.13]	0.000
Spatialiser - Verbaliser	Visual	0.710	[-0.07, 1.49]	0.083
Visualiser - Spatialiser	Spatial	0.045	[-0.66, 0.75]	0.988
Visualiser - Verbaliser	Spatial	0.331	[-0.23, 0.89]	0.347
Spatialiser - Verbaliser	Spatial	0.286	[-0.43, 1]	0.614
Visualiser - Spatialiser	Verbal	-0.388	[-1.07, 0.3]	0.381
Visualiser - Verbaliser	Verbal	-0.404	[-0.95, 0.14]	0.186
Spatialiser - Verbaliser	Verbal	-0.016	[-0.71, 0.67]	0.998
Visualiser - Spatialiser	Semantic	0.146	[-0.73, 1.02]	0.919
Visualiser - Verbaliser	Semantic	-0.041	[-0.71, 0.63]	0.989
Spatialiser - Verbaliser	Semantic	-0.187	[-1.07, 0.7]	0.874
Visualiser - Spatialiser	Sensorimotor	0.104	[-0.61, 0.82]	0.937
Visualiser - Verbaliser	Sensorimotor	0.267	[-0.3, 0.83]	0.508
Spatialiser - Verbaliser	Sensorimotor	0.163	[-0.56, 0.89]	0.857

```
m_strats_osivq |> report_contrast(~ strategy | cluster) |> knitr::kable()
```

Contrast	Cluster	Difference	95% CI	p.value
Visual - Spatial	Visualiser	0.297	[-0.33, 0.92]	0.699
Visual - Verbal	Visualiser	-0.331	[-0.95, 0.29]	0.590
Visual - Semantic	Visualiser	1.542	[0.84, 2.24]	0.000
Visual - Sensorimotor	Visualiser	0.582	[-0.05, 1.21]	0.087
Spatial - Verbal	Visualiser	-0.628	[-1.26, 0]	0.050
Spatial - Semantic	Visualiser	1.245	[0.54, 1.95]	0.000
Spatial - Sensorimotor	Visualiser	0.285	[-0.35, 0.92]	0.740
Verbal - Semantic	Visualiser	1.873	[1.17, 2.58]	0.000
Verbal - Sensorimotor	Visualiser	0.913	[0.28, 1.55]	0.001
Semantic - Sensorimotor	Visualiser	-0.960	[-1.67, -0.25]	0.002
Visual - Spatial	Spatialiser	-0.469	[-1.46, 0.52]	0.697
Visual - Verbal	Spatialiser	-1.529	[-2.52, -0.54]	0.000
Visual - Semantic	Spatialiser	0.877	[-0.25, 2]	0.207
Visual - Sensorimotor	Spatialiser	-0.124	[-1.12, 0.87]	0.997
Spatial - Verbal	Spatialiser	-1.061	[-2.02, -0.1]	0.021
Spatial - Semantic	Spatialiser	1.346	[0.24, 2.45]	0.008
Spatial - Sensorimotor	Spatialiser	0.344	[-0.63, 1.32]	0.871
Verbal - Semantic	Spatialiser	2.406	[1.3, 3.51]	0.000
Verbal - Sensorimotor	Spatialiser	1.405	[0.44, 2.37]	0.001
Semantic - Sensorimotor	Spatialiser	-1.002	[-2.11, 0.11]	0.099
Visual - Spatial	Verbaliser	-0.893	[-1.62, -0.16]	0.007
Visual - Verbal	Verbaliser	-2.256	[-2.99, -1.52]	0.000
Visual - Semantic	Verbaliser	-0.020	[-0.81, 0.77]	1.000
Visual - Sensorimotor	Verbaliser	-0.672	[-1.4, 0.06]	0.090
Spatial - Verbal	Verbaliser	-1.363	[-2.02, -0.7]	0.000
Spatial - Semantic	Verbaliser	0.873	[0.14, 1.61]	0.010
Spatial - Sensorimotor	Verbaliser	0.221	[-0.45, 0.89]	0.896
Verbal - Semantic	Verbaliser	2.236	[1.5, 2.97]	0.000
Verbal - Sensorimotor	Verbaliser	1.584	[0.91, 2.25]	0.000
Semantic - Sensorimotor	Verbaliser	-0.652	[-1.39, 0.08]	0.111

```
m_strats_osivq |> report_contrast(~ cluster * strategy, interaction = TRUE) |>
  knitr::kable()
```

Cluster contrast	Strategy contrast	Difference	95% CI	p.value
Visualiser - Spatialiser	Visual - Spatial	0.765	[-0.08, 1.61]	0.075
Visualiser - Verbaliser	Visual - Spatial	1.190	[0.5, 1.88]	0.001
Spatialiser - Verbaliser	Visual - Spatial	0.424	[-0.46, 1.31]	0.346
Visualiser - Spatialiser	Visual - Verbal	1.198	[0.36, 2.03]	0.005
Visualiser - Verbaliser	Visual - Verbal	1.925	[1.24, 2.61]	0.000
Spatialiser - Verbaliser	Visual - Verbal	0.727	[-0.14, 1.6]	0.102
Visualiser - Spatialiser	Visual - Semantic	0.665	[-0.28, 1.61]	0.170
Visualiser - Verbaliser	Visual - Semantic	1.562	[0.8, 2.32]	0.000
Spatialiser - Verbaliser	Visual - Semantic	0.897	[-0.09, 1.88]	0.075
Visualiser - Spatialiser	Visual - Sensorimotor	0.706	[-0.14, 1.55]	0.103
Visualiser - Verbaliser	Visual - Sensorimotor	1.254	[0.56, 1.95]	0.000
Spatialiser - Verbaliser	Visual - Sensorimotor	0.547	[-0.34, 1.44]	0.227
Visualiser - Spatialiser	Spatial - Verbal	0.433	[-0.39, 1.25]	0.301

Cluster contrast	Strategy contrast	Difference	95% CI	p.value
Visualiser - Verbaliser	Spatial - Verbal	0.735	[0.08, 1.39]	0.027
Spatialiser - Verbaliser	Spatial - Verbal	0.302	[-0.53, 1.13]	0.475
Visualiser - Spatialiser	Spatial - Semantic	-0.101	[-1.04, 0.84]	0.833
Visualiser - Verbaliser	Spatial - Semantic	0.372	[-0.36, 1.1]	0.317
Spatialiser - Verbaliser	Spatial - Semantic	0.473	[-0.48, 1.42]	0.329
Visualiser - Spatialiser	Spatial - Sensorimotor	-0.059	[-0.89, 0.78]	0.890
Visualiser - Verbaliser	Spatial - Sensorimotor	0.064	[-0.6, 0.73]	0.850
Spatialiser - Verbaliser	Spatial - Sensorimotor	0.123	[-0.73, 0.97]	0.776
Visualiser - Spatialiser	Verbal - Semantic	-0.534	[-1.46, 0.4]	0.260
Visualiser - Verbaliser	Verbal - Semantic	-0.363	[-1.08, 0.36]	0.322
Spatialiser - Verbaliser	Verbal - Semantic	0.170	[-0.77, 1.11]	0.722
Visualiser - Spatialiser	Verbal - Sensorimotor	-0.492	[-1.32, 0.33]	0.243
Visualiser - Verbaliser	Verbal - Sensorimotor	-0.671	[-1.33, -0.02]	0.044
Spatialiser - Verbaliser	Verbal - Sensorimotor	-0.180	[-1.02, 0.66]	0.674
Visualiser - Spatialiser	Semantic - Sensorimotor	0.042	[-0.9, 0.98]	0.931
Visualiser - Verbaliser	Semantic - Sensorimotor	-0.308	[-1.04, 0.42]	0.409
Spatialiser - Verbaliser	Semantic - Sensorimotor	-0.350	[-1.3, 0.61]	0.473

Visualisation

Strategy data were visualised using two different representations: as average Likert scores and as proportions of answers in each Likert category for each strategy and group. These figures were created with the functions `plot_strategies_scores()` (for the average scores) and `plot_strategies_barplot()` (for the proportions) from the package. A little `add_significance()` helper function was also created to add significance stars to the average score plots.

```
library(patchwork)

star_size <- 3

ps1 <-
  plot_strategies_scores(
    df_strats_long,
    grouping = group_2,
    title = "VVIQ 2 groups",
    plot.background = ggplot2::element_rect(fill = "white", colour = NA)
  ) +
  add_significance(
    size_star = star_size,
    color = "#1b6096",
    tibble::tibble(
      x_star = 1.28,
      y_star = 4.67,
      stars = "***",
      x_line = .data$x_star - 0.5,
      x_line_end = .data$x_star + 0.5,
      y_line = 4.6
    )
  )

ps2 <-
```



```

plot_strategies_scores(
  df_strats_long,
  grouping = group_3,
  title = "VVIQ 3 groups",
  plot.background = ggplot2::element_rect(fill = "white", colour = NA)
) +
add_significance(
  size_star = star_size,
  color = "#1b6096",
  tibble::tibble(
    x_star = 1.28,
    y_star = 4.67,
    stars = "**",
    x_line = .data$x_star - 0.5,
    x_line_end = .data$x_star + 0.5,
    y_line = 4.6
  )
) +
add_significance(
  size_star = star_size,
  color = "#1b6096",
  tibble::tibble(
    x_star = 2.28,
    y_star = 4.79,
    stars = "**",
    x_line = .data$x_star - 0.5,
    x_line_end = .data$x_star + 0.5,
    y_line = 4.72
  )
) +
add_significance(
  size_star = star_size,
  color = "#1b6096",
  tibble::tibble(
    x_star = 1.78,
    y_star = 5.07,
    stars = "***",
    x_line = .data$x_star - 1,
    x_line_end = .data$x_star + 1,
    y_line = 5
  )
) +
add_significance(
  size_star = star_size,
  color = "#318f2c",
  linetype = "longdash",
  lw = 0.3,
  tibble::tibble(
    x_star = 2.38,
    y_star = 4.54,
    stars = "*",
    x_line = .data$x_star - 0.5,
    x_line_end = .data$x_star + 0.5,

```

```

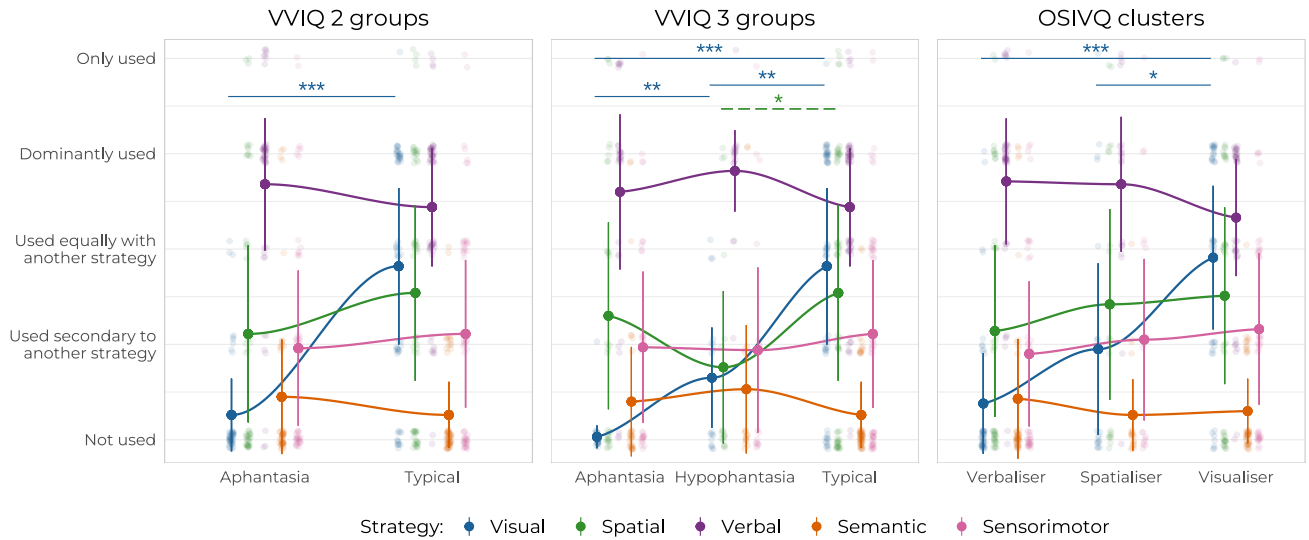
    y_line = 4.47
  )
)

ps3 <-
  plot_strategies_scores(
    df_strats_long,
    grouping = cluster,
    title = "OSIVQ clusters",
    plot.background = ggplot2::element_rect(fill = "white", colour = NA)
  ) +
  add_significance(
    size_star = star_size,
    color = "#1b6096",
    tibble::tibble(
      x_star = 2.28,
      y_star = 4.79,
      stars = "*",
      x_line = .data$x_star - 0.5,
      x_line_end = .data$x_star + 0.5,
      y_line = 4.72
    )
  ) +
  add_significance(
    size_star = star_size,
    color = "#1b6096",
    tibble::tibble(
      x_star = 1.78,
      y_star = 5.07,
      stars = "***",
      x_line = .data$x_star - 1,
      x_line_end = .data$x_star + 1,
      y_line = 5
    )
  )
)

ps <-
  ps1 + ps2 + ps3 +
  patchwork::plot_layout(axes = "collect", guides = "collect") &
  ggplot2::theme(legend.position = "bottom")

plot(ps)

```



```
pb1 <-
  plot_strategies_barplot(
    df_strats_long, group_2, title = "VVIQ 2 groups",
    plot.background = ggplot2::element_rect(fill = "white", colour = NA)
  )
pb2 <-
  plot_strategies_barplot(
    df_strats_long, group_3, title = "VVIQ 3 groups",
    plot.background = ggplot2::element_rect(fill = "white", colour = NA)
  )
pb3 <-
  plot_strategies_barplot(
    df_strats_long, cluster, title = "OSIVQ clusters",
    plot.background = ggplot2::element_rect(fill = "white", colour = NA)
  )

pb <-
  pb1 + pb2 + pb3 +
  patchwork::plot_layout(guides = "collect") &
  ggplot2::theme(legend.position = "bottom")

plot(pb)
```



All done!

```

#> - Session info -----
#> setting value
#> version R version 4.5.0 (2025-04-11 ucrt)
#> os      Windows 11 x64 (build 26100)
#> system  x86_64, mingw32
#> ui      RTerm
#> language (EN)
#> collate French_France.utf8
#> ctype   French_France.utf8
#> tz      Europe/Paris
#> date    2025-09-17
#> pandoc  3.6.3 @ C:/Program Files/RStudio/resources/app/bin/quarto/bin/tools/ (via rmarkdown)
#>
#> - Packages -----
#> ! package          * version      date (UTC) lib source
#> P abind             1.4-8        2024-09-12 [?] CRAN (R 4.5.0)
#> P aphantasiaReasoningViie * 0.0.0.9000  2025-09-16 [1] local
#> P assertthat        0.2.1        2019-03-21 [?] CRAN (R 4.5.0)
#> P class             7.3-23       2025-01-01 [3] CRAN (R 4.5.0)
#> P cli               3.6.5        2025-04-23 [?] CRAN (R 4.5.0)
#> P clue              0.3-66       2024-11-13 [?] CRAN (R 4.5.0)
#> P cluster           2.1.8.1      2025-03-12 [3] CRAN (R 4.5.0)
#> P clusterCrit       1.3.0        2023-11-23 [?] CRAN (R 4.5.0)
#> P clValid           0.7          2021-02-14 [?] CRAN (R 4.5.0)
#> P coda              0.19-4.1     2024-01-31 [?] CRAN (R 4.5.0)
#> P combinat          0.0-8        2012-10-29 [?] CRAN (R 4.5.0)
#> P crayon            1.5.3        2024-06-20 [?] CRAN (R 4.5.0)
#> P curl              6.3.0        2025-06-06 [?] RSPM
#> P diceR             3.1.0        2025-06-19 [?] RSPM (R 4.5.0)
#> P digest            0.6.37       2024-08-19 [?] CRAN (R 4.5.0)
#> P dplyr             1.1.4        2023-11-17 [?] CRAN (R 4.5.0)
#> P e1071             1.7-16       2024-09-16 [?] CRAN (R 4.5.0)
#> P emmeans           1.11.1       2025-05-04 [?] CRAN (R 4.5.0)
#> P estimability      1.5.1        2024-05-12 [?] CRAN (R 4.5.0)
#> P evaluate          1.0.4        2025-06-18 [?] RSPM (R 4.5.0)
#> P farver            2.1.2        2024-05-13 [?] CRAN (R 4.5.0)
#> P fastmap           1.2.0        2024-05-15 [?] CRAN (R 4.5.0)
#> P forcats           1.0.0        2023-01-29 [?] CRAN (R 4.5.0)
#> P generics          0.1.4        2025-05-09 [?] CRAN (R 4.5.0)
#> P ggplot2           3.5.2        2025-04-09 [?] CRAN (R 4.5.0)
#> P glue              1.8.0        2024-09-30 [?] CRAN (R 4.5.0)
#> P gtable            0.3.6        2024-10-25 [?] CRAN (R 4.5.0)
#> P haven             2.5.5        2025-05-30 [?] RSPM (R 4.5.0)
#> P highr             0.11         2024-05-26 [?] CRAN (R 4.5.0)
#> P hms               1.1.3        2023-03-21 [?] CRAN (R 4.5.0)
#> P htmltools         0.5.8.1      2024-04-04 [?] CRAN (R 4.5.0)
#> P httpuv            1.6.16       2025-04-16 [?] CRAN (R 4.5.0)
#> P insight           1.3.1        2025-06-30 [?] RSPM (R 4.5.0)
#> P jsonlite          2.0.0        2025-03-27 [?] CRAN (R 4.5.0)
#> P klaR              1.7-3        2023-12-13 [?] CRAN (R 4.5.0)
#> P knitr             1.50         2025-03-16 [?] CRAN (R 4.5.0)
#> P labeling          0.4.3        2023-08-29 [?] CRAN (R 4.5.0)
#> P labelled          2.14.1       2025-05-06 [?] CRAN (R 4.5.0)
#> P later             1.4.2        2025-04-08 [?] CRAN (R 4.5.0)

```

```

#> P lattice 0.22-7 2025-04-02 [?] CRAN (R 4.5.0)
#> P lifecycle 1.0.4 2023-11-07 [?] CRAN (R 4.5.0)
#> P magrittr 2.0.3 2022-03-30 [?] CRAN (R 4.5.0)
#> P MASS 7.3-65 2025-02-28 [3] CRAN (R 4.5.0)
#> P Matrix 1.7-3 2025-03-11 [3] CRAN (R 4.5.0)
#> P mclust 6.1.1 2024-04-29 [?] CRAN (R 4.5.0)
#> P mgcv 1.9-3 2025-04-04 [?] CRAN (R 4.5.0)
#> P mime 0.13 2025-03-17 [?] CRAN (R 4.5.0)
#> P miniUI 0.1.2 2025-04-17 [?] CRAN (R 4.5.0)
#> P mvtnorm 1.3-3 2025-01-10 [?] CRAN (R 4.5.0)
#> P nlme 3.1-168 2025-03-31 [3] CRAN (R 4.5.0)
#> P numDeriv 2016.8-1.1 2019-06-06 [?] CRAN (R 4.5.0)
#> P ordinal 2023.12-4.1 2024-08-19 [?] CRAN (R 4.5.0)
#> P patchwork * 1.3.1 2025-06-21 [?] RSPM (R 4.5.0)
#> P performance 0.14.0 2025-05-22 [?] CRAN (R 4.5.0)
#> P pillar 1.10.2 2025-04-05 [?] CRAN (R 4.5.0)
#> P pkgconfig 2.0.3 2019-09-22 [?] CRAN (R 4.5.0)
#> P promises 1.3.3 2025-05-29 [?] RSPM
#> P proxy 0.4-27 2022-06-09 [?] CRAN (R 4.5.0)
#> P purrr 1.0.4 2025-02-05 [?] CRAN (R 4.5.0)
#> P questionr 0.8.1 2025-06-10 [?] RSPM (R 4.5.0)
#> P R6 2.6.1 2025-02-15 [?] CRAN (R 4.5.0)
#> P RColorBrewer 1.1-3 2022-04-03 [?] CRAN (R 4.5.0)
#> P Rcpp 1.0.14 2025-01-12 [?] CRAN (R 4.5.0)
#> P rlang 1.1.6 2025-04-11 [?] CRAN (R 4.5.0)
#> P rmarkdown 2.29 2024-11-04 [?] CRAN (R 4.5.0)
#> P rstudioapi 0.17.1 2024-10-22 [?] CRAN (R 4.5.0)
#> P scales 1.4.0 2025-04-24 [?] CRAN (R 4.5.0)
#> P sessioninfo 1.2.2 2021-12-06 [?] RSPM (R 4.5.0)
#> P shiny 1.10.0 2024-12-14 [?] CRAN (R 4.5.0)
#> P showtext 0.9-7 2024-03-02 [?] CRAN (R 4.5.0)
#> P showtextdb 3.0 2020-06-04 [?] CRAN (R 4.5.0)
#> P stringi 1.8.7 2025-03-27 [?] CRAN (R 4.5.0)
#> P stringr 1.5.1 2023-11-14 [?] CRAN (R 4.5.0)
#> P sysfonts 0.8.9 2024-03-02 [?] CRAN (R 4.5.0)
#> P tibble 3.3.0 2025-06-08 [?] RSPM
#> P tidyr 1.3.1 2024-01-24 [?] CRAN (R 4.5.0)
#> P tidyselect 1.2.1 2024-03-11 [?] CRAN (R 4.5.0)
#> P ucminf 1.2.2 2024-06-24 [?] CRAN (R 4.5.0)
#> P vctrs 0.6.5 2023-12-01 [?] CRAN (R 4.5.0)
#> P viridisLite 0.4.2 2023-05-02 [?] CRAN (R 4.5.0)
#> P withr 3.0.2 2024-10-28 [?] CRAN (R 4.5.0)
#> P xfun 0.52 2025-04-02 [?] CRAN (R 4.5.0)
#> P xtable 1.8-4 2019-04-21 [?] CRAN (R 4.5.0)
#> P yaml 2.3.10 2024-07-26 [?] CRAN (R 4.5.0)
#>
#> [1] C:/Users/Mael/AppData/Local/R/cache/R/renv/library/aphantasiaReasoningViie-d318fbb0/windows/R-4
#> [2] C:/Users/Mael/AppData/Local/R/cache/R/renv/sandbox/windows/R-4.5/x86_64-w64-mingw32/9bc714e4
#> [3] C:/Program Files/R/R-4.5.0/library
#>
#> P -- Loaded and on-disk path mismatch.
#>
#> -----

```